

Lab: Connected -- Reading Minds and Sending Signals

Objective

Learning Goal: Explore how communication works as shared prediction -- modeling other people's minds and aligning your mental models with theirs.

This is a **group challenge** designed for middle school students (Grades 6-8).

Materials Needed

- Paper and pencils (for drawing)
 - A partner or small group
 - Timer or stopwatch
 - Lab journal or notebook
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Part 1: The Telephone Challenge (10 min)

Communication fails when mental models do not align. Let's see this in action.

Round 1 -- Describe and Draw: One person describes a simple image (without showing it) while their partner tries to draw it based only on the verbal description. No peeking.

Round 2 -- Same image, better description: Try again with the same image but this time the drawer can ask YES/NO questions.

Round	Accuracy (1-10)	How many questions asked?	What caused the biggest misunderstanding?
1 (no questions)		0	
2 (with questions)			

Write your response here

Part 2: Social Mind-Reading (15 min)

In Active Inference, communication is about modeling other people's mental states. You do this every day -- predicting what someone means, how they feel, and what they will say next.

Emoji Translation Challenge: For each text message below, write what you think the sender ACTUALLY means and how confident you are.

Text Message	What they literally said	What you think they MEAN	Confidence (1-10)
"Fine."			
"lol sure"			
"We should hang out sometime"			
"No worries!"			
"K"			

Now compare your interpretations with a partner. Did you agree?

Write your response here

Part 3: Communication Breakdown Analysis (10 min)

Think of a time when a miscommunication happened to you or someone you know. Map out what went wrong using the Active Inference framework.

Element	What Happened
What did the sender intend to communicate?	
What did the receiver actually understand?	
Where was the prediction error?	
What prior beliefs caused the misunderstanding?	
How was it (or could it have been) resolved?	

Write your response here

Part 4: Persuasion and Influence (5 min)

Communication is not just sharing information -- it is also about changing other people's mental models. That is what persuasion is.

Think about a time you tried to convince someone of something (a parent, a friend, a teacher).
Fill in:

Step	Your Example
What were you trying to convince them of?	
What was their prior belief (before you tried)?	
What evidence or argument did you use?	
Did their belief update? Why or why not?	

Write your response here

Reflection Table

Question	Your Answer
What does it mean to "model someone else's mind"?	
Why does two-way communication work better than one-way?	
How do prior beliefs affect how we interpret messages?	
What is the Active Inference view of communication?	
How can understanding prediction help you communicate better?	

Write your response here